# Problem Set XII, Macroeconomics III
# University of Copenhagen

## Exercise 1: Optimal monetary policy in Sweden

The Riksbank is Sweden's central bank. Like other central banks, it has expanded its balance sheet in the past decade by purchasing financial assets. When asset prices fall, the Riksbank loses money (its portfolio value decreases); when prices rise, it makes a profit.

To curb inflation, the Riksbank, like other central banks, has used contractionary monetary policy. This has not only helped stabilize or lower everyday prices but has also reduced the value of the Riksbank's assets, resulting in losses.

Unlike some central banks, the Riksbank must be recapitalized by the Swedish parliament if it incurs losses, meaning it needs to ask for government funds. This process is a humbling and politically sensitive process for the central bank, which it likely wants to avoid.

In this exercise, we study the consequences of the political cost of recapitalization. Assume that the Riksbank's loss function is given by

$$\mathcal{L}(\pi_t, x_t) = \frac{1}{2}\left[\alpha(\pi_t - \overline{\pi})^2 + \lambda(x_t - \overline{x})^2 - \gamma m_t\right]$$

where $\pi_t$ is the current inflation rate in Sweden, $\overline{\pi} > 0$ is the Riksbank's inflation target, $x_t$ is current output, $\overline{x}$ is the Riksbank's output target and $m_t$ is the money supply, controlled by the Riksbank. The parameters $\alpha$, $\lambda$ and $\gamma$ govern the severity of the bank's utility loss due to inflationary deviations from target, output deviations from target, and money supply changes, respectively. Money supply $m_t$ enters negatively, because the bank is averse to lowering the money supply: decreasing the $m_t$ lowers the value of other assets in the economy (including the assets held by the Riksbank), which makes recapitalization more likely – this hurts the central bank.

Assume that current output is described by

$$x_t = \theta_t + \pi_t - \pi_t^e$$

where $\theta_t$ is potential output. The value of $\theta$ is known to all agents (central bank and private sector) before expectations are formed and economic actions are taken each period. Then, $\pi_t^e$ is the private sector's expected rate of inflation, given $\theta_t$. Potential output $\theta_t$ is random, with realizations drawn from a normal distribution with mean zero and variance $\sigma_\theta$.

Finally, the central bank sets the money supply $m_t$ after the private sector has formed expectations. The realized rate of inflation is $\pi_t = m_t$.

a) Assume that the central bank commits to a rule of the form $m_t = \phi + \phi_\theta \theta_t$ before the economy starts. Determine the optimal value of $\phi$ and $\phi_\theta$, chosen by the bank, before the economy starts and any shocks have realized. Provide expressions for the inflation rate and output under this rule, $\pi^C$ and $x^C$. How do $\alpha$ and $\gamma$ affect each of these realizations? Provide an economic intuition for your result.

b) The European central bank (ECB) has no contemporary recapitalization requirements, and hence no political cost of changing the money supply. What would $\pi_{EU}^C$ and $x_{EU}^C$ be if the European central bank committed to the same rule as the Riksbank in (a)?

c) For this exercise, set the inflation target to $\overline{\pi} = 0$. Now, assume that the Riksbank cannot credibly commit to a policy rule. Thus, it minimizes its loss function in a discretionary way, setting the money supply after all shocks have realized each period and the private sector has formed expectations. Show that the realized rate of inflation is $\pi_t^D = \frac{1}{2}\frac{\gamma}{\alpha} + \frac{\lambda}{\alpha}(\overline{x} - \theta_t)$. Provide an intuition for your result.

d) What is realized output in Sweden, $x_t$, under the discretionary policy? Explain.

e) Recently, a new governor was appointed to lead the Riksbank. He was widely described as an inflation "hawk", meaning that the private sector expected him to be very averse to inflation. What does this information imply for the model? Is the Riksbank's potential aversion to recapitalization a problem?

# Exercise 2: Optimal monetary policy

Assume that the economy can be described as follows: the central bank can directly set the rate of inflation $\pi_t$. It faces a loss function of the form

$$\mathcal{L}(\pi_t, x_t) = \frac{1}{2}\left[\pi_t^2 + \lambda(x_t - \overline{x})^2\right]$$

where $x_t = \theta_t + \pi_t - \pi_t^e$. The variable $x_t$ describes output in the economy, $\overline{x}$ is society's preferred level of output, $\theta_t$ is potential output and $\pi_t^e$ is the private sector's expectation of the rate of inflation; $\lambda > 0$ governs the central bank's preference for a small output gap.

Potential output $\theta_t$ is random. Each period, its realization is drawn from a normal distribution with mean $0$ and variance $\sigma_\theta$. The realizations are independent across time and known to all actors at the beginning of each period. Hence, $\theta_t$ is known to the private sector before they form their inflation expectations $\pi_t^e$. The central bank sets the rate of inflation after the private sectors expectations are formed. All actors in the model form rational expectations.

a) Assume that the central bank can commit to a monetary policy rule of the form $\pi_t = \phi + \phi_\theta \theta_t$ before the economy starts. Determine the optimal values of $\phi$ and $\phi_\theta$ and show that the realized values of inflation and output are $\pi^c = 0$ and $x^c = \theta_t$, respectively. [*Hint 1*: Since the rule is set before any shock realizations, the optimum needs to be found in expectations. *Hint 2*: Remember that $\theta_t$ is 0 in expectation, is independent of all other variables in the economy and that $E[\theta_t^2] = \sigma_\theta^2$]

b) Give an intuition for the fact that $\lambda$ does not enter the solutions for $\pi^c$ and $x^c$.

c) Now assume that the central bank cannot credibly commit to a monetary policy rule and instead sets the value of $\pi_t$ each period under discretion. Show that as a result, the rate of inflation becomes $\pi_t^D = \lambda(\overline{x} - \theta_t)$. Explain why, under this policy, the inflation rate changes, while output stays at its potential level, with $x_t^D = \theta_t$. [*Hint*: Remember that, each period, this is a two-stage game, with the central bank moving last, taking expectations as given.]

d) Assume that the economy is in the discretionary equilibrium (as in Question c), implying that agents do not expect the central bank to adhere to any policy rule. Inflation expectations, hence, will be $\pi_t^e = \lambda(\overline{x} - \theta_t)$, once $\theta_t$ is known. In this environment, compute the central bank's **expected** losses in period $t$ under two scenarios (i.e., before $\theta_t$ is observed):

a.1) The central bank sets $\pi_t = \pi^D$, implementing the discretionary solution [*Hint*: $\mathbb{E}[\mathcal{L}(\pi^D, x^D)]$]

a.2) The central bank sets $\pi_t = \pi^C$, unexpectedly implementing the commitment solution

What is the lower bound for the value of relative future benefits of implementing option (b) that makes the central bank indifferent between the two options? [*Hint 1:* Remember that benefits in this context imply a smaller loss function. *Hint 2:* You do not need to solve the infinite horizon problem to answer this question. ]